

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Single and Multivariate Calculus | Course Code: 25EN1201 | Credits: 4

Exam Session: End Semester Exam May 2026 (Max Marks: 80, Duration: 2h 30m) Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Find the maximum and minimum values of the function $f(x, y) = 2x + y$ subject to the constraint $x^2 + y^2 = 5$ using the method of Lagrange Multipliers. [8 Marks]

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Q2. Verify Green's Theorem in the plane for the vector field $F = (xy + y^2) \mathbf{i} + x^2 \mathbf{j}$ around the closed curve bounded by $y = x$ and $y = x^2$. [8 Marks]

MODULE 2

Q3. Apply Gauss Divergence Theorem to compute the flux of $F = 4xz \mathbf{i} - y^2 \mathbf{j} + yz \mathbf{k}$ across the surface bounded by the cylinder $x^2 + y^2 = 4$ and planes $z = 0, z = 3$. [8 Marks]

--- OR ---

Q4. Test the convergence of the infinite series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$ ($x > 0$) using D'Alembert's Ratio Test. [6 Marks]